1)

$\frac{1}{r(r+1)} = \frac{A}{r} - \frac{B}{r+1}$

$1 = A(r+1) - Br$

When $r = -1$, $1 = B$

When $r = 0$, $1 = A$

$\therefore A = 1, B = 1$

$$\sum_{r=1}^{n} \frac{1}{r(r+1)} = \sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{r+1}\right) =$$

$$\begin{array}{ll} \frac{1}{1} & -\frac{1}{2} \\ +\frac{1}{2} & -\frac{1}{3} \\ +\frac{1}{3} & -\frac{1}{4} \\ \ldots & \ldots \\ +\frac{1}{n-2} & -\frac{1}{n-1} \\ +\frac{1}{n-1} & -\frac{1}{n} \\ +\frac{1}{n} & -\frac{1}{n+1} \end{array}$$

$= 1 - \frac{1}{n+1} = \frac{n}{n+1}$

2)

$\frac{1}{r(r+1)(r+2)} = \frac{A}{r} + \frac{B}{r+1} + \frac{C}{r+2}$

$1 = A(r+1)(r+2) + Br(r+2) + Cr(r+1)$

When $r = 0$, $1 = 2A$

When $r = -1$, $1 = -B$

When $r = -2$, $1 = 2C$

$\therefore A = \frac{1}{2}, B = -1$ and $C = \frac{1}{2}$

$$\sum_{r=1}^{40} \frac{1}{r(r+1)(r+2)} = \sum_{r=1}^{40} \left(\frac{1}{2r} - \frac{1}{r+1} + \frac{1}{2(r+2)}\right) =$$

$$\frac{1}{2} \quad -\frac{1}{2} \quad +\frac{1}{6}$$

$$+\frac{1}{4} \quad -\frac{1}{3} \quad +\frac{1}{8}$$

$$+\frac{1}{6} \quad -\frac{1}{4} \quad +\frac{1}{10}$$

$$\cdots \quad \cdots \quad \cdots$$

$$+\frac{1}{76} \quad -\frac{1}{39} \quad +\frac{1}{80}$$

$$+\frac{1}{78} \quad -\frac{1}{40} \quad +\frac{1}{82}$$

$$+\frac{1}{80} \quad -\frac{1}{41} \quad +\frac{1}{84}$$

$= \frac{1}{2} - \frac{1}{2} + \frac{1}{4} + \frac{1}{82} - \frac{1}{41} + \frac{1}{84} = \frac{215}{861}$

3)

$\frac{1}{r^2+5r+6} = \frac{1}{(r+2)(r+3)} = \frac{A}{r+2} + \frac{B}{r+3}$

$1 = A(r+3) + B(r+2)$

When $r = -3$, $1 = -B$

When $r = -2$, $1 = A$

$\therefore A = 1, B = -1$

$\frac{1}{(r+2)(r+3)} = \frac{1}{r+2} - \frac{1}{r+3}$

$$\sum_{r=10}^{200} \frac{1}{(r+2)(r+3)} = \sum_{r=10}^{200} \left(\frac{1}{r+2} - \frac{1}{r+3}\right) =$$

$$\begin{array}{ll} \frac{1}{12} & -\frac{1}{13} \\ +\frac{1}{13} & -\frac{1}{14} \\ +\frac{1}{14} & -\frac{1}{15} \\ \cdots & \cdots \\ +\frac{1}{200} & -\frac{1}{201} \\ +\frac{1}{201} & -\frac{1}{202} \\ +\frac{1}{202} & -\frac{1}{203} \end{array}$$

$= \frac{1}{12} - \frac{1}{203}$

$$\therefore \sum_{r=10}^{200} \frac{1}{(r+2)(r+3)} = \frac{191}{2436}$$

4)

$\frac{1}{(r+1)(r+2)(r+3)} = \frac{A}{r+1} + \frac{B}{r+2} + \frac{C}{r+3}$

$1 = A(r+2)(r+3) + B(r+1)(r+3) + C(r+1)(r+2)$

When $r = -1$, $1 = 2A$

When $r = -2$, $1 = -B$

When $r = -3$, $1 = 2C$

$\therefore A = \frac{1}{2}, B = -1, C = \frac{1}{2}$

$$\sum_{r=5}^{55} \frac{1}{(r+1)(r+2)(r+3)} = \sum_{r=5}^{55} \left(\frac{1}{2(r+1)} - \frac{1}{r+2} + \frac{1}{2(r+3)} \right) =$$

$$\frac{1}{12} \quad -\frac{1}{7} \quad +\frac{1}{16}$$
$$+\frac{1}{14} \quad -\frac{1}{8} \quad +\frac{1}{18}$$
$$+\frac{1}{16} \quad -\frac{1}{9} \quad +\frac{1}{20}$$
$$\cdots \quad \cdots \quad \cdots$$
$$+\frac{1}{108} \quad -\frac{1}{55} \quad +\frac{1}{112}$$
$$+\frac{1}{110} \quad -\frac{1}{56} \quad +\frac{1}{114}$$
$$+\frac{1}{112} \quad -\frac{1}{57} \quad +\frac{1}{116}$$

$= \frac{1}{12} - \frac{1}{7} + \frac{1}{14} + \frac{1}{114} - \frac{1}{57} + \frac{1}{116} = \frac{136}{11571}$

5)

$\frac{4}{(r+3)(r+4)} = \frac{A}{r+3} + \frac{B}{r+4}$

$4 = A(r+4) + B(r+3)$

When $r = -4$, $4 = -B$

When $r = -3$, $4 = 4A$

$\therefore A = 1, B = -1$

$\frac{4}{(r+3)(r+4)} = \frac{1}{r+3} - \frac{1}{r+4}$

$$\sum_{r=10}^{n} \frac{4}{(r+3)(r+4)} = \sum_{r=10}^{n} \left(\frac{1}{r+3} - \frac{1}{r+4}\right) =$$

$$\frac{1}{13} - \frac{1}{14}$$
$$+\frac{1}{14} - \frac{1}{15}$$
$$+\frac{1}{15} - \frac{1}{16}$$
$$\cdots \quad \cdots$$
$$+\frac{1}{n+1} - \frac{1}{n+2}$$
$$+\frac{1}{n+2} - \frac{1}{n+3}$$
$$+\frac{1}{n+3} - \frac{1}{n+4}$$

$= \frac{1}{13} - \frac{1}{n+4}$

$= \frac{n+4}{13(n+4)} - \frac{13}{13(n+4)}$

$= \frac{n-9}{13(n+4)}$

6)

$4r^2 - 4r - 3 = (2r + 1)(2r - 3)$

$\frac{4}{4r^2-4r-3} = \frac{4}{(2r+1)(2r-3)} = \frac{A}{2r+1} + \frac{B}{2r-3}$

$4 = A(2r - 3) + B(2r + 1)$

When $r = \frac{3}{2}$, $4 = 4B$

When $r = -\frac{1}{2}$, $4 = -4A$

$\therefore A = -1, B = 1$

$\frac{4}{4r^2-4r-3} = \frac{1}{2r-3} - \frac{1}{2r+1}$

$$\sum_{r=2}^{n} \frac{4}{4r^2 - 4r - 3} = \sum_{r=2}^{n} \left(\frac{1}{2r - 3} - \frac{1}{2r + 1}\right) =$$

$$\begin{array}{rr}
\frac{1}{1} & -\frac{1}{5} \\
+\frac{1}{3} & -\frac{1}{7} \\
+\frac{1}{5} & -\frac{1}{9} \\
\cdots & \cdots \\
+\frac{1}{2n-7} & -\frac{1}{2n-3} \\
+\frac{1}{2n-5} & -\frac{1}{2n-1} \\
+\frac{1}{2n-3} & -\frac{1}{2n+1}
\end{array}$$

$= \frac{4}{3} - \frac{1}{2n-1} - \frac{1}{2n+1}$

$= \frac{4}{3} - \frac{2n+1}{(2n-1)(2n+1)} - \frac{2n-1}{(2n-1)(2n+1)}$

$= \frac{4(4n^2-1)}{3(2n-1)(2n+1)} - \frac{4n}{(2n-1)(2n+1)}$

$= \frac{4(4n^2-1)}{3(2n-1)(2n+1)} - \frac{12n}{3(2n-1)(2n+1)}$

$= \frac{4(4n^2-3n-1)}{3(4n^2-1)}$

$= \frac{4(n-1)(4n+1)}{12n^2-3}$

7) a)

$r^2 + 4r + 3 = (r + 3)(r + 1)$

$= \frac{2}{(r+3)(r+1)} = \frac{A}{r+3} + \frac{B}{r+1}$

$2 = A(r + 1) + B(r + 3)$

When $r = -1$, $2 = 2B$

When $r = -3$, $2 = -2A$

$\therefore A = -1, B = 1$

$\frac{2}{r^2+4r+3} = \frac{1}{r+1} - \frac{1}{r+3}$

$$\sum_{r=1}^{n} \frac{2}{r^2 + 4r + 3} = \sum_{r=1}^{n} \left(\frac{1}{r + 1} - \frac{1}{r + 3}\right) =$$

$$\begin{array}{rr} \frac{1}{2} & -\frac{1}{4} \\ +\frac{1}{3} & -\frac{1}{5} \\ +\frac{1}{4} & -\frac{1}{6} \\ \cdots & \cdots \\ +\frac{1}{n-1} & -\frac{1}{2n-3} \\ +\frac{1}{n} & -\frac{1}{n+2} \\ +\frac{1}{n+1} & -\frac{1}{n+3} \end{array}$$

$= \frac{1}{2} + \frac{1}{3} - \frac{1}{n+2} - \frac{1}{n+3}$

$= \frac{5}{6} - \frac{1}{n+2} - \frac{1}{n+3}$

7) b)

$$\sum_{r=1}^{n} \frac{1}{r^2 + 4r + 3} = \frac{1}{2}\left[\frac{5}{6} - \frac{1}{n + 2} - \frac{1}{n + 3}\right]$$

As $n \to \infty, \frac{1}{n+2} \to 0, \frac{1}{n+3} \to 0$

$$\sum_{r=1}^{\infty} \frac{1}{r^2 + 4r + 3} = \frac{5}{12}$$

8)

$\frac{1}{(2r+1)(2r+3)} = \frac{A}{2r+1} + \frac{B}{2r+3}$

$1 = A(2r+3) + B(2r+1)$

When $r = -\frac{3}{2}$, $1 = -2B$

When $r = -\frac{1}{2}$, $1 = 2A$

$\therefore A = \frac{1}{2}, B = -\frac{1}{2}$

$\frac{1}{(2r+1)(2r+3)} = \frac{1}{2(2r+1)} - \frac{1}{2(2r+3)}$

$$\sum_{r=1}^{n} \frac{1}{(2r+1)(2r+3)} = \sum_{r=1}^{n} \left(\frac{1}{2(2r+1)} - \frac{1}{2(2r+3)} \right) =$$

$$\begin{array}{ll} \frac{1}{2\times3} & -\frac{1}{2\times5} \\ +\frac{1}{2\times5} & -\frac{1}{2\times7} \\ +\frac{1}{2\times7} & -\frac{1}{2\times9} \\ \cdots & \cdots \\ +\frac{1}{2(2n-3)} & -\frac{1}{2(2n-1)} \\ +\frac{1}{2(2n-1)} & -\frac{1}{2(2n+1)} \\ +\frac{1}{2(2n+1)} & -\frac{1}{2(2n+3)} \end{array}$$

$= \frac{1}{6} - \frac{1}{2(2n+3)}$

$$\therefore \sum_{r=1}^{\infty} \frac{1}{(2r+1)(2r+3)} = \frac{1}{6}$$

9) a)

$\frac{3}{r^2+3r} = \frac{3}{r(r+3)} = \frac{A}{r} + \frac{B}{r+3}$

$3 = A(r+3) + Br$

When $r = -3$, $1 = -1B$

When $r = 0$, $1 = 1A$

$\therefore A = 1, B = -1$

$\frac{3}{r^2+3r} = \frac{1}{r} - \frac{1}{r+3}$

$$\sum_{r=1}^{n} \frac{3}{r^2+3r} = \sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{r+3}\right) =$$

$$\begin{aligned}
&\frac{1}{1} && -\frac{1}{4}\\
&+\frac{1}{2} && -\frac{1}{5}\\
&+\frac{1}{3} && -\frac{1}{6}\\
&+\frac{1}{4} && -\frac{1}{7}\\
&\cdots && \cdots\\
&+\frac{1}{n-3} && -\frac{1}{n}\\
&+\frac{1}{n-2} && -\frac{1}{n+1}\\
&+\frac{1}{n-1} && -\frac{1}{n+2}\\
&+\frac{1}{n} && -\frac{1}{n+3}
\end{aligned}$$

$= 1 + \frac{1}{2} + \frac{1}{3} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{1}{n+3} = \frac{11}{6} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{1}{n+3}$

$$\sum_{r=1}^{n} \frac{1}{r^2+3r} = \frac{1}{3}\sum_{r=1}^{n} \frac{3}{r^2+3r} = \frac{1}{3}\left(\frac{11}{6} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{1}{n+3}\right)$$

9) b)

As $n \to \infty, \frac{1}{n+1} \to 0, \frac{1}{n+2} \to 0, \frac{1}{n+3} \to 0$

$$\sum_{r=1}^{\infty} \frac{1}{r^2+3r} = \frac{11}{18}$$

10)

$\frac{1}{r^2-3r+2} = \frac{1}{(r-1)(r-2)} = \frac{A}{r-1} + \frac{B}{r-2}$

$1 = A(r-2) + B(r-1)$

When $r = 1$, $1 = -A$

When $r = 2$, $1 = B$

$\therefore A = -1, B = 1$

$\frac{1}{r^2-3r+2} = \frac{1}{r-2} - \frac{1}{r-1}$

$$\sum_{r=3}^{n} \frac{1}{r^2-3r+2} = \sum_{r=3}^{n} \left(\frac{1}{r-2} - \frac{1}{r-1}\right)$$

So since

$$\sum_{r=3}^{n} \left(\frac{1}{r-2} - \frac{1}{r-1}\right) =$$

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n-4} + \frac{1}{n-3} + \frac{1}{n-2} - \frac{1}{2} - \frac{1}{3} - \frac{1}{4} - \cdots - \frac{1}{n-3} - \frac{1}{n-2} - \frac{1}{n-1}$$

$= 1 - \frac{1}{n-1}$

$\therefore 1 - \frac{1}{n-1} > 0.45 \Rightarrow \frac{1}{n-1} < 0.55$

Since the smallest value that n can be is $n = 3$ then $n - 1 > 0$ and so

$1 < 0.55(n-1) \Rightarrow 1 < 0.55n - 0.55$

$\therefore n > \frac{1.55}{0.55}$

So $n = 3$

11) a)

$\frac{1}{(3r-2)(3r+1)} = \frac{A}{3r-2} + \frac{B}{3r+1}$

$1 = A(3r+1) + B(3r-2)$

When $r = \frac{2}{3}$, $1 = 3A$

When $r = -\frac{1}{3}$, $1 = -3B$

$\therefore A = \frac{1}{3}, B = -\frac{1}{3}$

$\frac{1}{(3r-2)(3r+1)} = \frac{1}{3(3r-2)} - \frac{1}{3(3r+1)}$

$$\sum_{r=n}^{2n} \frac{1}{(3r-2)(3r+1)} = \sum_{r=n}^{2n} \left(\frac{1}{3(3r-2)} - \frac{1}{3(3r+1)} \right)$$

So since

$$\sum_{r=n}^{2n} \left(\frac{1}{3(3r-2)} - \frac{1}{3(3r+1)} \right) =$$

$\frac{1}{3(3n-2)}$ $-\frac{1}{3(3n+1)}$

$+\frac{1}{3(3n+1)}$ $-\frac{1}{3(3n+4)}$

$+\frac{1}{3(3n+4)}$ $-\frac{1}{3(3n+7)}$

… …

$+\frac{1}{3(6n-8)}$ $-\frac{1}{3(6n-5)}$

$+\frac{1}{3(6n-5)}$ $-\frac{1}{3(6n-2)}$

$+\frac{1}{3(6n-2)}$ $-\frac{1}{3(6n+1)}$

$= \frac{1}{3(3n-2)} - \frac{1}{3(6n+1)}$

$= \frac{(6n+1)-(3n-2)}{3(3n-2)(6n+1)}$

$= \frac{3n+3}{3(3n-2)(6n+1)}$

$= \frac{n+1}{(3n-2)(6n+1)}$

11) b)

$$\lim_{n\to\infty}\left(\sum_{r=n}^{2n}\frac{1}{(3r-2)(3r+1)}\right)=\lim_{n\to\infty}\left(\frac{n+1}{(3n-2)(6n+1)}\right)$$

$$=\lim_{n\to\infty}\left(\frac{n+1}{18n^2-9n-2}\right)=\lim_{n\to\infty}\left(\frac{\frac{1}{n}+\frac{1}{n^2}}{18-\frac{9}{n}-\frac{2}{n^2}}\right)=0$$

12) a)

$\frac{15}{r^2+6r+8} = \frac{15}{(r+2)(r+4)} = \frac{A}{r+2} + \frac{B}{r+4}$

$15 = A(r+4) + B(r+2)$

When $r = -4$, $15 = -2B$

When $r = -2$, $15 = 2A$

$\therefore A = \frac{15}{2}, B = -\frac{15}{2}$

$\frac{15}{r^2+6r+8} = \frac{15}{2(r+2)} - \frac{15}{2(r+4)}$

$$\sum_{r=1}^{n} \frac{15}{r^2+6r+8} = \frac{15}{2}\sum_{r=1}^{n}\left(\frac{1}{r+2} - \frac{1}{r+4}\right)$$

So since

$$\sum_{r=1}^{n}\left(\frac{1}{r+2} - \frac{1}{r+4}\right) =$$

$$\begin{array}{ll} \frac{1}{3} & -\frac{1}{5} \\ +\frac{1}{4} & -\frac{1}{6} \\ +\frac{1}{5} & -\frac{1}{7} \\ \ldots & \ldots \\ +\frac{1}{n} & -\frac{1}{n+2} \\ +\frac{1}{n+1} & -\frac{1}{n+3} \\ +\frac{1}{n+2} & -\frac{1}{n+4} \end{array}$$

$= \frac{1}{3} + \frac{1}{4} - \frac{1}{n+3} - \frac{1}{n+4} = \frac{7}{12} - \frac{1}{n+3} - \frac{1}{n+4}$

$$\sum_{r=1}^{n} \frac{15}{r^2+6r+8} = \frac{15}{2}\sum_{r=1}^{n}\left(\frac{1}{r+2} - \frac{1}{r+4}\right) = \frac{15}{2}\left[\frac{7}{12} - \frac{1}{n+3} - \frac{1}{n+4}\right]$$

$= \frac{35}{8} - \frac{15}{2(n+3)} - \frac{15}{2(n+4)}$

$= \frac{35(n+3)(n+4)}{8(n+3)(n+4)} - \frac{60(n+4)}{8(n+3)(n+4)} - \frac{60(n+3)}{8(n+3)(n+4)}$

$= \frac{35n^2+245n+420-60n-240-60n-180}{8(n+3)(n+4)} = \frac{35n^2+125n}{8(n+3)(n+4)}$

12) b)

As $n \to \infty$, $\frac{15}{2(n+3)} \to 0$ and $\frac{15}{2(n+4)} \to 0$

$$\therefore \sum_{r=1}^{\infty} \frac{15}{r^2+6r+8} = \frac{35}{8}$$

So that there is 4.375 million litres of water left in the long term

12) c)

$$\sum_{r=1}^{30} 1 = 30 \Rightarrow \frac{35}{8} \times \sum_{r=1}^{30} \frac{1}{30} = \frac{35}{8} \Rightarrow \sum_{r=1}^{30} \frac{7}{48} = \frac{35}{8}$$

$$\therefore \sum_{r=1}^{n} \frac{15}{r^2+6r+8} - \sum_{r=1}^{30} \frac{7}{48}$$

13) a)

$\frac{300}{(r+1)(r+4)} = \frac{A}{r+1} + \frac{B}{r+4}$

$300 = A(r+4) + B(r+1)$

When $r = -1$, $300 = 3A$

When $r = -4$, $300 = -3B$

$\therefore A = 100, B = -100$

$\frac{300}{(r+1)(r+4)} = \frac{100}{r+1} - \frac{100}{r+4}$

$$\sum_{r=1}^{n} \frac{300}{(r+1)(r+4)} = 100 \sum_{r=1}^{n} \left(\frac{1}{r+1} - \frac{1}{r+4}\right)$$

So since

$$\sum_{r=1}^{n} \left(\frac{1}{r+1} - \frac{1}{r+4}\right) =$$

$$\begin{array}{ll} \frac{1}{2} & -\frac{1}{5} \\ +\frac{1}{3} & -\frac{1}{6} \\ +\frac{1}{4} & -\frac{1}{7} \\ \cdots & \cdots \\ +\frac{1}{n-1} & -\frac{1}{n+2} \\ +\frac{1}{n} & -\frac{1}{n+3} \\ +\frac{1}{n+1} & -\frac{1}{n+4} \end{array}$$

$= \frac{1}{2} + \frac{1}{3} + \frac{1}{4} - \frac{1}{n+2} - \frac{1}{n+3} - \frac{1}{n+4} = \frac{13}{12} - \frac{1}{n+2} - \frac{1}{n+3} - \frac{1}{n+4}$

$$\sum_{r=1}^{n} \frac{300}{(r+1)(r+4)} = 100 \sum_{r=1}^{n} \left(\frac{1}{r+1} - \frac{1}{r+4}\right) = 100\left[\frac{13}{12} - \frac{1}{n+2} - \frac{1}{n+3} - \frac{1}{n+4}\right]$$

$= \frac{1300}{12} - \frac{100}{n+2} - \frac{100}{n+3} - \frac{100}{n+4}$

$= \frac{1300(n+2)(n+3)(n+4)}{12(n+2)(n+3)(n+4)} - \frac{1200(n+3)(n+4)}{12(n+2)(n+3)(n+4)} - \frac{1200(n+2)(n+4)}{12(n+2)(n+3)(n+4)} - \frac{1200(n+2)(n+3)}{12(n+2)(n+3)(n+4)}$

$= \frac{25n(13n^2+81n+122)}{3(n+2)(n+3)(n+4)}$

13) b)

As $n \to \infty$, $\frac{1}{n+2} \to 0$, $\frac{1}{n+3} \to 0$ and $\frac{1}{n+4} \to 0$

$$\therefore \sum_{r=1}^{\infty} \frac{300}{(r+1)(r+4)} = 100 \times \frac{13}{12} = \frac{325}{3} = 108.3 \ldots$$

So the sum in the long term will never exceed the £109 million as claimed.

13) c)

$A_{\infty} = 100 \times \frac{13}{12} = \frac{325}{3}$ million

When $n = 30$:

$100\left[\frac{13}{12} - \frac{1}{32} - \frac{1}{33} - \frac{1}{34}\right] = \frac{445\,375}{4488} = 99.2$ million

Since $\frac{\frac{445375}{4488}}{\frac{325}{3}} \times 100 = 91.6\% > 90\%$ then the claim is verified that at least 90% of the total funding has been released.

14) a)

Since $\sin(A - B) = \sin A \cos B - \cos A \sin B$ then

$\sin r \cos(r+1) - \cos r \sin(r+1) = \sin\big(r - (r+1)\big) = \sin(-1)$

So

$$C_r = \frac{\sin(1)}{\cos(r)\cos(r+1)} = \frac{-(\sin(r)\cos(r+1) - \cos(r)\sin(r+1))}{\cos(r)\cos(r+1)}$$

$$= \frac{-\sin(r)\cos(r+1) + \cos(r)\sin(r+1)}{\cos(r)\cos(r+1)}$$

$$= -\frac{\sin(r)\cos(r+1)}{\cos(r)\cos(r+1)} + \frac{\cos(r)\sin(r+1)}{\cos(r)\cos(r+1)}$$

$$= -\tan(r) + \tan(r+1)$$

$= \tan(r+1) - \tan(r)$ as req.

14) b)

So

$C_r = \frac{\sin(-1)}{\cos r \sin(r+1)} = \tan(r+1) - \tan r$

$\therefore \sum_{r=1}^{n} C_r = \sum_{r=1}^{n} (\tan(r+1) - \tan r)$

$\sum_{r=1}^{n} (\tan(r+1) - \tan r) =$

~~$\tan 2$~~	$-\tan 1$
~~$+\tan 3$~~	~~$-\tan 2$~~
~~$+\tan 4$~~	~~$-\tan 3$~~
...	...
~~$+\tan(n-1)$~~	~~$-\tan(n-2)$~~
~~$+\tan n$~~	~~$-\tan(n-1)$~~
$+\tan(n+1)$	~~$-\tan n$~~

$= \tan(n+1) - \tan 1$

$\therefore \sum_{r=1}^{n} C_r = \sum_{r=1}^{n} (\tan(r+1) - \tan r) = \tan(n+1) - \tan 1$

15) a)

Since $\sinh(A-B) = \sinh A \cosh B - \cosh A \sinh B$ then

$\sinh(r+1)\cosh r - \cosh(r+1)\sinh r = \sinh 1$

So

$D_r = \frac{\sinh(1)}{\sinh(r)\sinh(r+1)} = \frac{\sinh(r+1)\cosh(r) - \cosh(r+1)\sinh(r)}{\sinh(r)\sinh(r+1)}$

$= \frac{\sinh(r+1)\cosh(r)}{\sinh(r)\sinh(r+1)} - \frac{\cosh(r+1)\sinh(r)}{\sinh(r)\sinh(r+1)}$

$= \coth(r) - \coth(r+1)$

15) b)

So

$$D_r = \frac{\sinh 1}{\sinh(r)\sinh(r+1)} = \coth r - \coth(r+1)$$

$$\therefore \sum_{r=1}^{n} D_r = \sum_{r=1}^{n} (\coth r - \coth(r+1))$$

$$\sum_{r=1}^{n} (\coth r - \coth(r+1)) =$$

$$\begin{array}{ll} \coth 1 & -\coth 2 \\ +\coth 2 & -\coth 3 \\ +\coth 3 & -\coth 4 \\ \ldots & \ldots \\ +\coth(n-2) & -\coth(n-1) \\ +\coth(n-1) & -\coth n \\ +\coth n & -\coth(n+1) \end{array}$$

$$= \coth 1 - \coth(n+1)$$

$$\therefore \sum_{r=1}^{n} D_r = \sum_{r=1}^{n} (\coth r - \coth(r+1)) = \coth 1 - \coth(n+1)$$

16) a)

$\tan\big((r+1)x - rx\big) = \tan x$ and

$$\tan\big((r+1)x - rx\big) = \frac{\tan\big((r+1)x\big) - \tan(rx)}{1 + \tan\big((r+1)x\big)\tan(rx)}$$

So that

$$\tan x = \frac{\tan\big((r+1)x\big) - \tan(rx)}{1 + \tan\big((r+1)x\big)\tan(rx)}$$

$$1 + \tan\big((r+1)x\big)\tan(rx) = \frac{\tan\big((r+1)x\big) - \tan(rx)}{\tan x}$$

$$\tan\big((r+1)x\big)\tan(rx) = \frac{\tan\big((r+1)x\big)}{\tan x} - \frac{\tan(rx)}{\tan x} - 1 \text{ as req.}$$

16) b)

Let $x = \frac{\pi}{20}$ so that

$$\tan\frac{\pi}{20}\tan\frac{2\pi}{20} + \tan\frac{2\pi}{20}\tan\frac{3\pi}{20} + \cdots + \tan\frac{14\pi}{20}\tan\frac{15\pi}{20}$$

$$= \sum_{r=1}^{14} \tan\left(\frac{\pi}{20}r\right)\tan\left((r+1)\frac{\pi}{20}\right) = \sum_{r=1}^{14}\left(\frac{\tan\left((r+1)\frac{\pi}{20}\right)}{\tan\frac{\pi}{20}} - \frac{\tan\left(\frac{\pi}{20}r\right)}{\tan\frac{\pi}{20}} - 1\right) =$$

$$\cancel{\frac{\tan\frac{2\pi}{20}}{\tan\frac{\pi}{20}}} \qquad -\cancel{\frac{\tan\frac{\pi}{20}}{\tan\frac{\pi}{20}}} \qquad -1$$

$$+\cancel{\frac{\tan\frac{3\pi}{20}}{\tan\frac{\pi}{20}}} \qquad -\cancel{\frac{\tan\frac{2\pi}{20}}{\tan\frac{\pi}{20}}} \qquad -1$$

$$+\cancel{\frac{\tan\frac{4\pi}{20}}{\tan\frac{\pi}{20}}} \qquad -\cancel{\frac{\tan\frac{3\pi}{20}}{\tan\frac{\pi}{20}}} \qquad -1$$

$$\cancel{\ldots} \qquad \cancel{\ldots} \qquad \ldots$$

$$+\cancel{\frac{\tan\frac{14\pi}{20}}{\tan\frac{\pi}{20}}} \qquad -\cancel{\frac{\tan\frac{13\pi}{20}}{\tan\frac{\pi}{20}}} \qquad -1$$

$$+\frac{\tan\frac{15\pi}{20}}{\tan\frac{\pi}{20}} \qquad -\cancel{\frac{\tan\frac{14\pi}{20}}{\tan\frac{\pi}{20}}} \qquad -1$$

$$= -\frac{\tan\frac{\pi}{20}}{\tan\frac{\pi}{20}} + \frac{\tan\frac{15\pi}{20}}{\tan\frac{\pi}{20}} - 14$$

$$= \frac{\tan\frac{15\pi}{20}}{\tan\frac{\pi}{20}} - 15$$

17) a)

$$u_{r+1} - u_r = \ln(r+2) - \ln(r+1) = \ln\left(\frac{r+2}{r+1}\right) = \ln\left(\frac{r+1+1}{r+1}\right) = \ln\left(1 + \frac{1}{r+1}\right)$$

17) b)

The mean value says that if f is continuous on $[a, b]$ then there exists $c \in (a, b)$ such that $\int_a^b \mathrm{f}(x)\,\mathrm{d}x = \mathrm{f}(c)(b-a)$

$$\int_{r+1}^{r+2} \frac{1}{x}\mathrm{d}x = [\ln x]_{r+1}^{r+2} = \ln(r+2) - \ln(r+1)$$

By the mean value theorem, there exists a $c \in (r+1, r+2)$ such that

$$\int_{r+1}^{r+2} \frac{1}{x}\mathrm{d}x = \frac{1}{c}\big((r+2) - (r+1)\big) = \frac{1}{c}$$

Hence

$u_{r+1} - u_r = \frac{1}{c}$ where $c \in (r+1, r+2)$

Since $c \in (r+1, r+2)$ then $r+1 < c < r+2$ and so $\frac{1}{r+2} < \frac{1}{c} < \frac{1}{r+1}$

$\therefore \frac{1}{r+2} < u_{r+1} - u_r < \frac{1}{r+1}$ as req.

17) c)

$$\sum_{r=1}^{n}(u_{r+1} - u_r) =$$

$+u_2$ $\quad -u_1$

$+u_3$ $\quad -u_2$

$+u_4$ $\quad -u_3$

$\ldots$ $\quad \ldots$

$+u_{n-1}$ $\quad -u_{n-2}$

$+u_n$ $\quad -u_{n-1}$

$+u_{n+1}$ $\quad -u_n$

$= u_{n+1} - u_1$

$= \ln(n+1+1) - \ln(1+1)$

$= \ln(n+2) - \ln(2)$

18) a)

$\frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$

$1 = A(x+1) + Bx$

When $x = 0$, $1 = A$

When $x = -1$, $1 = -B$

$\therefore A = 1, B = -1$

$\frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$

So $y = \int \left(\frac{1}{x} - \frac{1}{x+1}\right) \mathrm{d}x = \ln x - \ln(x+1) + c$

General solution is $y = \ln\left(\frac{x}{x+1}\right) + c$

18) b)

$u_{r+1} - u_r = \ln\left(\frac{r+1}{r+2}\right) - \ln\left(\frac{r}{r+1}\right)$

Alternatively:

$u_{r+1} - u_r = (\ln(r+1) - \ln(r+2) + c) - (\ln(r) - \ln(r+1) + c)$

$= -\ln(r) + 2\ln(r+1) - \ln(r+2)$

18) c)

$$\sum_{r=1}^{n}(u_{r+1}-u_r)=\sum_{r=1}^{n}\left(\ln\left(\frac{r+1}{r+2}\right)-\ln\left(\frac{r}{r+1}\right)\right)=$$

$$\ln\left(\tfrac{2}{3}\right) \qquad -\ln\left(\tfrac{1}{2}\right)$$
$$+\ln\left(\tfrac{3}{4}\right) \qquad -\ln\left(\tfrac{2}{3}\right)$$
$$+\ln\left(\tfrac{4}{5}\right) \qquad -\ln\left(\tfrac{3}{4}\right)$$
$$\dots \qquad \dots$$
$$+\ln\left(\tfrac{n-1}{n}\right) \qquad -\ln\left(\tfrac{n-2}{n-1}\right)$$
$$+\ln\left(\tfrac{n}{n+1}\right) \qquad -\ln\left(\tfrac{n-1}{n}\right)$$

$$=\ln\left(\tfrac{n+1}{n+2}\right)-\ln\left(\tfrac{1}{2}\right)$$
$$=\ln\left(\tfrac{2(n+1)}{n+2}\right)$$

Alternatively:

$$\sum_{r=1}^{n}(u_{r+1}-u_r)=\sum_{r=1}^{n}(-\ln(r)+2\ln(r+1)-\ln(r+2))=$$

$$-\ln 1 \qquad +2\ln 2 \qquad -\ln 3$$
$$-\ln 2 \qquad +2\ln 3 \qquad -\ln 4$$
$$-\ln 3 \qquad +2\ln 4 \qquad -\ln 5$$
$$\dots \qquad \dots \qquad \dots$$
$$+\ln(n-2) \qquad +2\ln(n-1) \qquad -\ln n$$
$$+\ln(n-1) \qquad +2\ln n \qquad -\ln(n+1)$$
$$+\ln n \qquad +2\ln(n+1) \qquad -\ln(n+2)$$

$$=-\ln 1+2\ln 2-\ln 2-\ln(n+1)+2\ln(n+1)-\ln(n+2)$$
$$=\ln 2+\ln(n+1)-\ln(n+2)$$
$$=\ln\left(\tfrac{2(n+1)}{n+2}\right)$$

18) d)

$$\sum_{r=1}^{\infty}(u_{r+1}-u_r)=\sum_{r=1}^{\infty}\left(\ln\left(\frac{2(n+1)}{n+2}\right)\right)=\ln 2$$